MODULE 3: FUNCTIONAL MATERIALS (CO: 3)

Complete Comprehensive Textbook: Band Theory, Semiconductors, Superconductors, Magnetic Materials, Nanomaterials & Solved VIT FAT Question Bank

Aligned with Official VIT BACHY105 Applied Chemistry Course Syllabus, Department Question Bank & VIT FAT Papers

Syllabus Scope & Competency Checklist (Module 3):

- **Electrical Materials:** Ohm's law ($J = \sigma E$), electrical resistivity (ρ), conductivity ($\sigma = ne\mu$).
- **Band Theory of Solids:** Overlap of atomic orbitals, valence band (VB), conduction band (CB), forbidden energy bandgap (E_g); classification into conductors ($E_g = 0$), semiconductors ($0 < E_g < 3.0$ eV), and insulators ($E_g > 5.0$ eV).
- **Semiconductors & Superconductors:** Intrinsic semiconductors ($n = p = n_i$); extrinsic semiconductors (n-type donor doping, p-type acceptor doping, Fermi level shifts); Superconductivity fundamentals, critical temperature (T_c), critical magnetic field ($H_{c(T)} = H_0\left[1 - \left(\frac{T}{T_c}\right)^2\right]$), zero electrical resistance, Meissner effect (perfect diamagnetism $\chi = -1, B = 0$), Type-I vs Type-II superconductors, BCS electron pairing (Cooper pairs).
- **Magnetic Materials:** Origins of magnetic moments (electron orbital and spin angular momentum); Diamagnetism ($\chi < 0$); Paramagnetism ($\chi > 0, \chi \propto \frac{1}{T}$ Curie law); Ferromagnetism (spontaneous magnetic domains, Weiss molecular field, Curie-Weiss law); Antiferromagnetism (sub-lattice antiparallel cancellation, Néel temperature T_N); Ferrimagnetism (ferrites, uncancelled antiparallel moments); B-H Hysteresis loop (Retentivity B_r, Coercivity H_c, magnetic domain wall pinning, hard vs soft magnetic materials for data storage).
- **Liquid Cooling & Thermal Fluids:** Heat transfer mechanisms, types of coolants (deionized water, ethylene glycol-water mixtures, dielectric mineral/fluorocarbon oils, liquid metals like Ga-In alloys), thermal conductivity (k), nanofluids for CPU/EV thermal management.
- **Nanomaterials & Quantum Dots:** Nanoscale regime ($1 - 100$ nm); high surface-area-to-volume ratio ($\frac{S}{V} \propto \frac{1}{r}$); Top-down synthesis (ball milling, photolithography, chemical etching) vs Bottom-up synthesis (sol-gel, chemical vapor deposition CVD, hydrothermal, coprecipitation); Bulk gold (inert, metallic yellow, melting point 1064°C) vs Nano gold (catalytically active, localized surface plasmon resonance LSPR ruby red color, melting point depression down to 300°C); Quantum confinement effect, exiton Bohr radius, Quantum Dots, Brus equation for size-dependent bandgap tuning.

1. Electrical Materials & Band Theory of Solids

1.1 Electrical Conduction & Ohm's Law

At the macroscopic level, electrical current density J is directly proportional to the applied electric field E:

$$J = \sigma E = \frac{1}{\rho} E$$

where σ is electrical conductivity ($\mathrm{S/m}$), ρ is electrical resistivity ($\Omega \cdot \mathrm{m}$), n is free charge carrier concentration, e is electronic charge (1.602×10^{-19} C), and μ is charge carrier mobility:

$$\sigma = ne\mu_e + pe\mu_h$$

1.2 Band Theory of Solids

When an enormous number of atoms ($N \approx 10^{23}$) assemble to form a crystalline solid lattice, their discrete atomic orbitals overlap and split into continuous allowed energy bands separated by forbidden energy gaps (**bandgaps, E_g**):

1. **Valence Band (VB):** The highest energy band occupied by valence electrons at 0 K.
2. **Conduction Band (CB):** The lowest unoccupied or partially filled energy band where excited electrons are free to drift under an electric field.
3. **Classification:**
   - **Metals (Conductors):** Conduction and valence bands overlap ($E_g = 0$), or the valence band is only partially filled (e.g., Cu, Al, Na). Thermal energy easily excites electrons into adjacent vacant states ($\sigma \approx 10^7$ S/m). As temperature rises, lattice phonon scattering increases, causing resistance to increase ($\rho \propto T$).
   - **Semiconductors:** Possess a narrow forbidden energy bandgap ($0.5 \leq E_g \leq 3.0$ eV) (e.g., Si with $E_g = 1.12$ eV, Ge with $E_g = 0.67$ eV, GaAs with $E_g = 1.42$ eV). At 0 K, the valence band is completely full and conduction band is completely empty (perfect insulator). Thermal energy at room temperature excites electrons across E_g, creating free electrons in CB and vacant holes in VB. Conductivity increases exponentially with temperature ($\sigma \propto e^{-\frac{E_g}{2k_B T}}$).

- **Insulators:** Possess an extremely wide forbidden bandgap ($E_g > 5.0$ eV) (e.g., diamond with $E_g = 5.47$ eV, quartz SiO_2 with $E_g = 9.0$ eV). Thermal excitation at normal temperatures is virtually zero ($\sigma < 10^{-12}$ S/m).

2. Semiconductors & The Physics of Superconductivity

2.1 Intrinsic vs Extrinsic Semiconductors

- **Intrinsic Semiconductors:** Pure elemental semiconductors (pure Si or Ge). Carrier generation is purely thermal:

$$n = p = n_i = \sqrt{N_c N_v} e^{-\frac{E_g}{2k_B T}}$$

The Fermi level (E_F) lies exactly at the center of the bandgap: $E_F = \frac{E_c + E_v}{2}$.

- **Extrinsic Semiconductors (Doped):**
  1. **N-Type (Donor Doped):** Doped with pentavalent impurities (P, As, Sb). Four valence electrons form covalent bonds; the fifth electron is loosely bound ($E_d \approx 0.05$ eV below CB). Majority carriers are electrons ($n \approx N_D \gg p$). The Fermi level shifts upward near the conduction band edge ($E_F \approx E_c - k_B T \ln\left(\frac{N_c}{N_D}\right)$).
  2. **P-Type (Acceptor Doped):** Doped with trivalent impurities (B, Al, Ga, In). Creates electron deficiency (holes) in the valence band ($E_a \approx 0.05$ eV above VB). Majority carriers are holes ($p \approx N_A \gg n$). The Fermi level shifts downward near the valence band edge ($E_F \approx E_v + k_B T \ln\left(\frac{N_v}{N_A}\right)$).

2.2 Superconductivity: Zero Resistance & The Meissner Effect

Discovered by Heike Kamerlingh Onnes (1911) in mercury at 4.2 K:

- **Zero Electrical Resistance:** Below a characteristic **Critical Temperature (T_c)**, electrical resistivity drops abruptly to zero ($\rho < 10^{-25} \Omega \cdot \text{m}$). Persistent currents induced in superconducting rings circulate indefinitely without attenuation.
- **Critical Magnetic Field (H_c):** Superconductivity is destroyed if an external magnetic field exceeds $H_{c(T)}$:

$$H_{c(T)} = H_0 \left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

- **The Meissner Effect (Perfect Diamagnetism):** When cooled below T_c in the presence of an external magnetic field H, a superconductor completely expels all magnetic flux from its interior ($B = 0$):

$$B = \mu_0(H + M) = 0 \implies M = -H \implies \chi = \frac{M}{H} = -1$$

  The magnetic susceptibility is identically -1. This diamagnetic expulsion produces stable magnetic levitation (Maglev trains).
- **BCS Theory (Cooper Pairs):** John Bardeen, Leon Cooper, and John Robert Schrieffer (1957) explained that below T_c, electron-lattice-electron interactions mediated by acoustic phonons overcome Coulomb repulsion, binding electrons of opposite momentum and spin into bosons called **Cooper Pairs** ($k\uparrow, -k\downarrow$). Cooper pairs condense into a macroscopic quantum ground state that flows through the crystal lattice without scattering.

3. Magnetic Materials & B-H Hysteresis Loops

3.1 Classification of Magnetic Materials

Magnetism originates from atomic magnetic dipole moments caused by electron orbital motion ($\boldsymbol{\mu}_l$) and intrinsic electron spin ($\boldsymbol{\mu}_s = -g\mu_B \boldsymbol{S}$).

1. **Diamagnetism:** Present in all atoms with completely filled electron shells (paired electrons, net $\boldsymbol{\mu} = 0$). External field induces opposing orbital currents (Lenz's law). Susceptibility is negative, small ($\chi \approx -10^{-5}$), and temperature-independent (e.g., Bi, Cu, Au, water).
2. **Paramagnetism:** Materials with permanent unpaired atomic magnetic dipole moments that are randomly oriented by thermal agitation. External field weakly aligns dipoles. Susceptibility is positive, small ($\chi \approx 10^{-4} - 10^{-2}$), and inversely proportional to temperature (**Curie's Law**):

$$\chi = \frac{C}{T}$$

3. **Ferromagnetism:** Materials (Fe, Co, Ni, NdFeB) exhibiting strong spontaneous magnetization below the **Curie Temperature (T_c)** even in zero applied field. Quantum mechanical **exchange interaction** aligns adjacent electron

spins parallel across microscopic domains ($10 - 100 \mu\text{m}$). Above T_c, thermal energy destroys domain alignment, transforming the material into a paramagnet following the **Curie-Weiss Law**:

$$\chi = \frac{C}{T - \theta}$$

4. **Antiferromagnetism:** Exchange interaction favors antiparallel alignment of adjacent atomic spins on interlocking sublattices, resulting in zero net macroscopic magnetic moment below the **Néel Temperature (T_N)** (e.g., MnO, FeO, Cr).
5. **Ferrimagnetism:** Antiparallel sublattices possess unequal magnetic moment magnitudes, yielding a net spontaneous magnetization (e.g., Ferrites Fe_3O_4, $\text{NiZnFe}_2\text{O}_4$). Highly insulating, preventing eddy current losses at high RF frequencies.

3.2 B-H Hysteresis Loop & Magnetic Data Storage

When an unmagnetized ferromagnetic specimen is subjected to an alternating magnetic field H:

- **Initial Magnetization Curve:** Magnetic domain walls displace easily, then domains rotate into alignment with H until saturation flux density B_s is reached.
- **Retentivity (Remanence B_r):** The residual magnetic flux density remaining in the material when the applied magnetizing field is reduced to zero ($H = 0$).
- **Coercivity (Coercive Field H_c):** The reverse magnetic field intensity required to completely demagnetize the material ($B = 0$).
- **Hysteresis Energy Loss:** The area enclosed by the B-H loop equals the energy dissipated as heat per unit volume per magnetization cycle:

$$\text{Loss} = \oint H dB$$

- **Hard vs Soft Magnetic Materials:**
  - **Soft Magnets:** Narrow hysteresis loop, high permeability, low coercivity ($H_c < 100 \text{ A/m}$), low hysteresis loss. Used for transformer cores, electric motor stators (e.g., Silicon steel, Permalloy).
  - **Hard Magnets:** Broad hysteresis loop, high retentivity ($B_r > 1.2 \text{ T}$), very high coercivity ($H_c > 10^5 \text{ A/m}$). Used for permanent magnets in BLDC motors and magnetic hard disk drive (HDD) data storage media where magnetic bits must resist accidental erasure.

4. Liquid Cooling & Nanomaterials: Quantum Dots & Brus Equation

4.1 Liquid Coolants & Thermal Management

Modern high-performance electronics (CPUs, GPUs, EV battery packs) generate heat fluxes exceeding 100 W/cm^2, demanding liquid cooling:

- **Cooling Fluids:**
  1. **Deionized Water:** Highest specific heat capacity ($c_p = 4184 \text{ J/(kg K)}$) and thermal conductivity ($k = 0.606 \text{ W/(m K)}$), but corrosive and electrically conductive if ionized.
  2. **Ethylene Glycol-Water Mixtures (50:50):** Prevents freezing down to $-37°\text{C}$ and raises boiling point; standard for automotive EV battery packs.
  3. **Dielectric Oils / Fluorinated Liquids (3M Novec):** Electrically insulating, permitting direct immersion cooling of server blades.
  4. **Nanofluids:** Stable suspensions of high-thermal-conductivity nanoparticles (Al_2O_3, CuO, carbon nanotubes) in base liquids, enhancing convective heat transfer coefficients by $15 - 30\%$.

4.2 Nanomaterials: Surface-to-Volume Ratio & Bulk vs Nano Gold

Nanomaterials are defined as substances possessing at least one dimension between 1 and 100 nanometers:

- **Surface Area-to-Volume Effect:** As particle radius r decreases:

$$\frac{\text{Surface Area}}{\text{Volume}} = \frac{4\pi r^2}{\frac{4}{3}\pi r^3} = \frac{3}{r}$$

A huge fraction of atoms reside at surface sites with unsaturated coordination bonds, drastically increasing surface free energy, chemical reactivity, and catalytic activity.

- **Dramatic Example: Bulk Gold vs Nanogold:**

- **Bulk Gold:** Inert noble metal, bright yellow metallic luster, unreactive, melting point $= 1064°\text{C}$.
- **Gold Nanoparticles ($10 - 20\ \text{nm}$):** Intense ruby-red / purple color due to **Localized Surface Plasmon Resonance (LSPR)** (coherent collective oscillation of conduction electrons in resonance with visible light at $520\ \text{nm}$); exceptionally active heterogeneous catalyst for low-temperature oxidation of toxic "CO" to CO_2; melting point decreases significantly down to $300 - 500°\text{C}$ due to high surface energy.

4.3 Quantum Confinement & The Brus Equation for Quantum Dots

When the physical size of a semiconductor crystal becomes smaller than its **Exciton Bohr Radius (r_B)**:

$$r_B = \frac{\varepsilon_r m_0}{\mu^*} a_0$$

where $\mu^* = \frac{m_e^* m_h^*}{m_e^* + m_h^*}$ is the reduced effective mass of the electron-hole pair. Charge carriers are spatially confined in three dimensions, converting continuous energy bands into discrete quantized atomic-like energy levels (**Quantum Dots, QDs**).

- **The Brus Equation:**

The size-dependent effective bandgap $E_g^*(R)$ of a spherical quantum dot of radius R is given by:

$$E_g^*(R) = E_g^{\text{bulk}} + \frac{h^2}{8\mu^* R^2} - \frac{1.8e^2}{4\pi\varepsilon_0\varepsilon_r R}$$

- **Physical Meaning of Terms:**
  1. E_g^{bulk}: The intrinsic macroscopic bulk semiconductor bandgap.
  2. $\frac{h^2}{8\mu^* R^2}$: The quantum confinement kinetic energy term (scales as $\frac{1}{R^2}$), which widens the bandgap as radius R shrinks.
  3. $-\frac{1.8e^2}{4\pi\varepsilon_0\varepsilon_r R}$: Coulombic electrostatic attraction between electron and hole (scales as $\frac{1}{R}$).
- **Technological Application:** By synthesizing CdSe or InP quantum dots with precise diameters from $2\ \text{nm}$ to $6\ \text{nm}$, photoluminescent emission can be tuned across the entire visible spectrum from bright blue to deep red without changing chemical composition. Utilized in high-gamut QLED television displays, bio-imaging fluorescent markers, and tandem solar cells.

VIT FAT Solved Question Bank: Step-by-Step Numerical Derivations

FAT Problem 1: Brus Equation & Bandgap Tuning in Semiconductor Quantum Dots
Question (10 Marks): A spherical CdSe semiconductor quantum dot has a radius of $R = 2.50$ nm. Given Constants:

- Bulk bandgap of CdSe: $E_g^{\text{bulk}} = 1.74$ eV (2.788×10^{-19} J)
- Electron effective mass: $m_e^* = 0.13 m_0$
- Hole effective mass: $m_h^* = 0.45 m_0$
- Free electron mass: $m_0 = 9.109 \times 10^{-31}$ kg
- Planck's constant: $h = 6.626 \times 10^{-34}$ J · s
- Relative dielectric constant: $\varepsilon_r = 10.6$
- Elementary charge: $e = 1.602 \times 10^{-19}$ C, $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

(“ ”) Calculate the reduced effective mass μ^* of the exciton. (“ ”) Compute the quantum confinement energy shift and the Coulombic attraction term. (“ ”) Determine the total effective bandgap E_g^* of the quantum dot in electron-volts (eV). (“ ”) Calculate the wavelength λ and color of light emitted during band-edge photoluminescence.

Complete Step-by-Step Solution:
Step 1: Calculate Reduced Effective Mass μ^*:

$$\mu^* = \frac{m_e^* m_h^*}{m_e^* + m_h^*} = \frac{0.13 \times 0.45}{0.13 + 0.45} m_0 = \frac{0.0585}{0.58} m_0 = 0.10086 m_0$$

$$\mu^* = 0.10086 \times (9.109 \times 10^{-31}\ \text{kg}) = 9.187 \times 10^{-32}\ \text{kg}$$

Step 2: Calculate Quantum Confinement Kinetic Energy:

$$\Delta E_{\text{conf}} = \frac{h^2}{8 \mu^* R^2}$$

$$h^2 = (6.626 \times 10^{-34})^2 = 4.3904 \times 10^{-67}\ \text{J}^2 \cdot \text{s}^2$$

$$R^2 = (2.50 \times 10^{-9}\ \text{m})^2 = 6.25 \times 10^{-18}\ \text{m}^2$$

$$8 \mu^* R^2 = 8 \times (9.187 \times 10^{-32}) \times (6.25 \times 10^{-18}) = 4.5935 \times 10^{-48}\ \text{kg} \cdot \text{m}^2$$

$$\Delta E_{\text{conf}} = \frac{4.3904 \times 10^{-67}}{4.5935 \times 10^{-48}} = 9.558 \times 10^{-20}\ \text{Joules}$$

In electron-volts:

$$\Delta E_{\text{conf}} = \frac{9.558 \times 10^{-20}}{1.602 \times 10^{-19}} = +0.5966\ \text{eV}$$

Step 3: Calculate Coulombic Attraction Energy:

$$\Delta E_{\text{Coul}} = \frac{1.8 e^2}{4 \pi \varepsilon_0 \varepsilon_r R}$$

$$1.8 e^2 = 1.8 \times (1.602 \times 10^{-19})^2 = 4.620 \times 10^{-38}\ \text{C}^2$$

$$4 \pi \varepsilon_0 \varepsilon_r R = (1.1126 \times 10^{-10}) \times 10.6 \times (2.5 \times 10^{-9}) = 2.948 \times 10^{-18}$$

$$\Delta E_{\text{Coul}} = \frac{4.620 \times 10^{-38}}{2.948 \times 10^{-18}} = 1.567 \times 10^{-20}\ \text{Joules} = 0.0978\ \text{eV}$$

Step 4: Total Effective Bandgap E_g^*:

$$E_g^* = E_g^{\text{bulk}} + \Delta E_{\text{conf}} - \Delta E_{\text{Coul}} = 1.74 + 0.5966 - 0.0978 = 2.2388\ \text{eV}$$

Step 5: Emission Wavelength:

$$\lambda = \frac{hc}{E_g^*} = \frac{1240\ \text{eV} \cdot \text{nm}}{2.2388\ \text{eV}} = 553.86\ \text{nm}$$

(Corresponds to brilliant yellow-green emission, whereas bulk CdSe at 1.74 eV emits invisible near-infrared at 713 nm).

Boxed Final Answers:

FAT Problem 2: Superconducting Critical Magnetic Field & Meissner Current Density
Question (10 Marks): A superconducting niobium wire has a critical temperature of $T_c = 9.25$ K in zero applied magnetic field. The critical magnetic field at absolute zero is $H_0 = 1.60 \times 10^5$ A/m. (“ ”) Calculate the critical magnetic field $H_{c(T)}$ of this wire at liquid helium boiling temperature ($T = 4.20$ K). (“ ”) Determine the maximum operating temperature at which the wire can sustain superconductivity in an external magnetic field of 1.0×10^5 A/m. (“ ”) If the wire diameter is $d = 1.50$ mm, compute the maximum critical electric current (I_c) that can pass through the wire at 4.20 K before superconductivity is destroyed (Silsbee’s rule).

Complete Step-by-Step Solution:
Step 1: Critical Field at 4.20 K using Tuyn’s Law:

$$H_{c(T)} = H_0 \left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$\frac{T}{T_c} = \frac{4.20}{9.25} = 0.45405$$

$$\left(\frac{T}{T_c}\right)^2 = (0.45405)^2 = 0.20616$$

$$H_{c(4.2\text{K})} = (1.60 \times 10^5) \times [1 - 0.20616] = (1.60 \times 10^5) \times 0.79384 = 1.270 \times 10^5 \text{ A/m}$$

Step 2: Maximum Operating Temperature for $H = 1.0 \times 10^5$ A/m:

$$1.0 \times 10^5 = 1.60 \times 10^5 \left[1 - \left(\frac{T}{9.25}\right)^2\right]$$

$$1 - \left(\frac{T}{9.25}\right)^2 = \frac{1.0 \times 10^5}{1.60 \times 10^5} = 0.625$$

$$\left(\frac{T}{9.25}\right)^2 = 1 - 0.625 = 0.375$$

$$\frac{T}{9.25} = \sqrt{0.375} = 0.61237 \implies T = 0.61237 \times 9.25 = 5.664 \text{ K}$$

Step 3: Maximum Critical Current (Silsbee’s Rule): By Silsbee’s hypothesis, the critical transport current I_c is that which creates a self-magnetic field at the wire surface equal to the critical magnetic field:

$$H_c = \frac{I_c}{2\pi r} = \frac{I_c}{\pi d} \implies I_c = \pi d H_c$$

Radius $r = 0.75$ mm $= 0.75 \times 10^{-3}$ m, Diameter $d = 1.50 \times 10^{-3}$ m:

$$I_c = \pi \times (1.50 \times 10^{-3} \text{ m}) \times (1.270 \times 10^5 \text{ A/m}) = 3.14159 \times 190.5 = 598.47 \text{ Amperes}$$

(The niobium wire can conduct an extraordinary ≈ 598.5 A without any resistance!).

Boxed Final Answers:

$$(?) H_{c(4.2\text{K})} = 1.27 \times 10^5 \text{ A/m}, \quad (?) T_{\max} = 5.66 \text{ K}, \quad (?) I_c = 598.5 \text{ A}.$$

FAT Problem 3: B-H Hysteresis Energy Loss & Transformer Core Sizing
Question (10 Marks): An electrical transformer core fabricated from soft silicon steel has a volume of $V = 0.012\ \text{m}^3$ and operates at a power grid frequency of $f = 50$ Hz. When cycled through magnetic saturation, the measured B-H hysteresis loop area corresponds to an energy loss of $280\ \text{Joules/m}^3$ per cycle. (“ ”) Calculate the hysteresis power loss dissipated in the core in Watts. (“ ”) If the core material has a density of $\rho = 7650\ \text{kg/m}^3$, calculate the hysteresis loss per unit mass (W/kg). (“ ”) If the operating frequency is increased to 400 Hz (aerospace power frequency) with the same flux density swing, calculate the new hysteresis power loss.

Complete Step-by-Step Solution:
Step 1: Hysteresis Power Loss at 50 Hz (P_h): The hysteresis energy lost per cycle per unit volume is $W_h = 280\ \text{J/m}^3$. Total power dissipated is:

$$P_h = W_h \times V \times f$$

$$P_h = (280\ \text{J/m}^3) \times (0.012\ \text{m}^3) \times (50\ \text{s}^{-1}) = 3.36 \times 50 = 168.0\ \text{Watts}$$

Step 2: Specific Loss Per Kilogram: Total core mass:

$$M_{\text{core}} = V \times \rho = 0.012\ \text{m}^3 \times 7650\ \text{kg/m}^3 = 91.80\ \text{kg}$$

Specific hysteresis loss:

$$\text{Specific Loss} = \frac{P_h}{M_{\text{core}}} = \frac{168.0}{91.80} = 1.830\ \text{W/kg}$$

Step 3: Power Loss at 400 Hz Frequency: Since hysteresis energy per cycle is constant, hysteresis power loss scales linearly with frequency ($P_h \propto f$):

$$P_{h(400\text{Hz})} = P_{h(50\text{Hz})} \times \left(\frac{400}{50}\right) = 168.0 \times 8 = 1344.0\ \text{Watts} = 1.344\ \text{kW}$$

Boxed Final Answers:

$$(\boxed{?})P_{h(50\text{Hz})} = 168.0\ \text{W}, \quad (\boxed{?})\ \text{Specific Loss} = 1.83\ \text{W/kg}, \quad (\boxed{?})P_{h(400\text{Hz})} = 1344.0\ \text{W}\ (1.34\ \text{kW}).$$